

STATE AI COMPLIANCE COST MODEL

Methodology · September 2026 · elliottjames-paschal.github.io/state-ai-compliance-cost

THE QUESTION

What does state-by-state AI regulation cost a company *beyond* what one federal standard would have cost? Existing figures come from macroeconomic models whose elasticity assumptions a reader cannot test. This model is built the opposite way: every assumption is a dashboard control, so a skeptic can substitute their own and see what their number is.

THE DATA

All 89 state AI laws passed in the 2026 session (82 distinct laws after companion-bill deduplication) were fetched as enacted texts and coded **twice, independently, by LLM** against a fixed protocol (requirement families, duty triggers, parameter vocabularies — `CODING.md` in the repo). The two passes materially agreed on 76 of 89 bills; the 13 disagreements were adjudicated from the statutory text. Every coded value is anchored to a verbatim quoted provision.

THE MODEL — BOTTOM-UP, HOURS × RATES

- **Duties drive cost.** Each law is coded as the duties it demands (AI disclosure, crisis protocol, impact assessment, ...). Each duty carries triangular hour estimates for the average AI model — legal review and engineering once, operations every year of the horizon.
- **Parameters drive discounts.** *Reuse* is the share of coded parameters (age thresholds, notice deadlines, audit cadence) two states set identically, averaged across state pairs in a family — that fraction of a build carries to the next state. The *federal baseline* is what every state sets identically: work a single national rule would require anyway, never counted, its size reported alongside the estimate. Both derive from the statutes; neither is assigned.
- **Sequencing.** Within each family, the first state pays its divergence from the federal baseline; each later state pays (1 – reuse) of a fresh build plus its recurring operations.
- **Range, not point.** 4,000 seeded Monte Carlo draws over the hour ranges, with a shared error factor across draws so uncertainty doesn't wash out in the sum. The conservative end (p10) leads.

SCOPE DISCIPLINE

Only laws that bind companies enter the estimate. Criminal prohibitions, state-agency rules, and school mandates are excluded by default — dashboard toggles show what indiscriminate counting would add. Duplicate companion bills are counted once. Statutory penalties for non-compliance are coded and displayed as fact, but never folded into the cost: doing so would require enforcement-probability assumptions of exactly the kind this model exists to avoid.

CAVEATS

Parameters, duties, and penalties are statutory facts with quoted sources. The per-duty hour defaults are informed estimates for the average AI model, editable on the dashboard and pending citation to published burden figures (Paperwork Reduction Act filings, state fiscal notes).